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RESEARCH ARTICLE

Deep Reinforcement Learning-Driven Dynamic Spectrum Access in Dense Wi-Fi Environments

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ABSTRACT The rapid growth of wireless devices and bandwidth-intensive applications in urban environments has exacerbated spectrum congestion in Wi-Fi networks, resulting in performance degradation in terms of latency, throughput, and fairness. Traditional mechanisms such as CSMA/CA and static channel allocation often fail to adapt effectively in high-density scenarios. To address this challenge, we propose a novel framework that leverages Deep Reinforcement Learning (DRL)—specifically Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO)—to enable intelligent, dynamic spectrum access in IEEE 802.11ax/be (Wi-Fi 6/7) environments. Using the NS-3 simulator integrated with PyTorch-based agents, we model dense deployments with multiple access points and varied traffic patterns, including VoIP, video, FTP, and AR/VR flows. Our DRL agents are trained to select frequency channels in real-time based on environmental observations, such as interference levels, traffic load, and delay, to maximize overall network quality of service. Comparative evaluations against legacy MAC schemes demonstrate that PPO improves average throughput by up to 38%, reduces end-to-end latency by 48%, and enhances fairness, while maintaining minimal switching overhead. Furthermore, we discuss the potential for prototyping the system using OpenWRT-enabled access points, demonstrating the framework's feasibility for real-world deployments in smart cities and enterprise networks.

INDEX TERMS Dynamic spectrum access, deep reinforcement learning, Wi-Fi 6, Wi-Fi 7, DQN, PPO, NS-3, channel allocation, urban wireless networks, OpenWRT.

I. INTRODUCTION

In recent years, the exponential growth of wireless devices and bandwidth-intensive applications has transformed urban environments into high-density wireless ecosystems. Wi-Fi networks, particularly those based on IEEE 802.11ax (Wi-Fi 6) and the emerging IEEE 802.11be (Wi-Fi 7) standards, are increasingly deployed in crowded venues such as apartment complexes, stadiums, and smart city hubs. These environments are characterized by spectrum scarcity, heavy interference, and intense contention for access, which severely degrade Quality of Service (QoS) metrics such as throughput, latency, and reliability [1], [6], [43]. Despite advances in modulation techniques and spectrum efficiency, the foundational contention-based Medium Access Control (MAC) mechanism—Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)—struggles to adapt in real time to dynamic interference patterns and fluctuating network loads [15], [26].

Static or semi-static channel assignment approaches are equally inadequate in such scenarios, often resulting in channel underutilization or severe co-channel interference [13], [24]. These limitations highlight the growing necessity for smarter, adaptive spectrum access mechanisms capable of learning from and responding to environmental changes in real time. With the proliferation of heterogeneous traffic patterns—ranging from high-throughput video conferencing to latency-sensitive augmented reality—there is an urgent need to move beyond legacy MAC designs toward more intelligent solutions [25], [30].

Deep Reinforcement Learning (DRL) has emerged as a powerful paradigm for sequential decision-making in

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complex wireless environments. Unlike traditional rule-based optimization schemes, DRL can autonomously learn optimal channel selection or spectrum access policies by interacting with the network and observing feedback over time [2], [5], [12], [24]. Prior work has explored DRL in various wireless contexts, including rate adaptation [13], link maintenance [3], edge computing resource allocation [5], and coexistence in unlicensed bands [19]. However, the application of DRL to dynamic spectrum access (DSA) in Wi-Fi 6/7 networks remains relatively underexplored, especially under high-density, real-world traffic conditions.

This study aims to bridge this gap by proposing a DRL-driven framework for dynamic spectrum access in dense Wi-Fi environments. Specifically, we develop and evaluate a Deep Q-Network (DQN) and Proximal Policy Optimization (PPO) based agent to dynamically select optimal channels based on real-time environmental cues such as signal-to-noise ratio (SNR), user density, and traffic load. The proposed system is integrated into an NS-3 simulation environment configured for IEEE 802.11ax/be and calibrated with realistic traffic patterns, including video streaming, VoIP, and AR/VR flows [13], [29], [33].

Our key contributions are as follows:

(i) We develop a DRL-based DSA framework tailored for Wi-Fi 6/7 networks under dense deployment conditions, leveraging DQN and PPO architectures for learning optimal channel policies [2], [12], [24];
(ii) We implement a high-fidelity simulation using NS-3, including various density scenarios and channel interference models, to validate the performance benefits over CSMA/CA and fixed channel assignment [33], [43];
(iii) We propose a prototype system architecture based on OpenWRT-enabled access points, outlining the practical feasibility of embedding the DRL agent into real-world deployments [8], [23], [36];
(iv) Finally, we conduct a comparative analysis against baseline Wi-Fi protocols to quantify improvements in throughput, latency, and spectrum utilization efficiency [13], [15], [26].

While prior studies have applied DRL to wireless networks, this work introduces several unique aspects that distinguish it from existing literature. First, unlike earlier DRL-based approaches in Wi-Fi that focused primarily on MAC parameter tuning, rate adaptation, or single-AP configurations, our framework explicitly addresses **dense multi-AP Wi-Fi 6/7 deployments** with dual-band operation and heterogeneous traffic mixes. Second, by integrating **NS-3 with PyTorch-based DRL agents**, we achieve a tightly coupled simulation-learning environment that enables reproducible benchmarking and realistic traffic modeling, which is rarely attempted in earlier work. Third, we extend beyond simulation by outlining a ** prototype architecture using OpenWRT-enabled access points**, thus bridging the gap between theoretical DRL design and practical implementation. Finally, this study incorporates a ** comprehensive evaluation** including convergence analysis, fairness assessment, and statistical significance tests, offering a more transparent and rigorous analysis compared to prior works. Together, these elements establish a clear advancement over existing DRL research in spectrum access. The rest of this paper is organized as follows: Section III discusses related work on DRL and Wi-Fi spectrum access. Section IV describes the system architecture and methodology. Section V presents the experimental setup and simulation parameters. Section VI analyzes the performance metrics. Section VII discusses practical implications and limitations, while Section VII concludes the paper with future directions.

II. BACKGROUND AND HISTORY

Traditional Wi-Fi networks have long relied on the Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) mechanism as their fundamental MAC protocol. While effective in sparse and moderately loaded environments, CSMA/CA suffers significant degradation in high-density deployments due to its contention-based nature, leading to increased collisions, backoffs, and access delays [15], [26], [43]. Even enhancements such as Enhanced Distributed Channel Access (EDCA), introduced in IEEE 802.11e to offer differentiated QoS, are inadequate under extreme congestion. EDCA's static access parameters and reliance on contention-based arbitration fail to scale well when numerous stations simultaneously contend for the medium [13], [30], [45].

Another traditional approach to mitigate interference and improve performance in dense Wi-Fi deployments involves static or heuristic-based channel selection. These techniques often rely on signal strength, interference measurements, or network heuristics to determine the optimal channel assignment for access points [9], [27], [33]. However, they typically operate in an offline or semi-static manner and cannot dynamically adapt to real-time variations in traffic load, user mobility, or interference. Furthermore, many of these approaches do not scale efficiently in multi-AP or dual-band environments, especially with emerging use cases like video conferencing, AR/VR, and edge-driven applications [6], [20], [34].

To overcome these limitations, researchers have increasingly explored machine learning techniques for wireless resource management, particularly in the domain of reinforcement learning (RL) and deep reinforcement learning (DRL). In LTE and 5G networks, DRL has been employed to optimize radio resource scheduling [2], enhance multi-access edge computing (MEC) performance [5], and improve network slicing efficiency [28], [30], [44]. In cognitive radio systems, DRL agents have demonstrated the ability to autonomously learn optimal spectrum access policies under dynamic and adversarial conditions, outperforming traditional Q-learning and heuristic models [19], [29], [48].

The use of DRL in Wi-Fi environments has also begun to gain momentum. Prior studies have investigated DRL-based approaches for rate adaptation [13], MAC parameter tuning [15], contention window optimization [26], and decentralized channel access [24], [40]. For instance, Cássia de et al. [15] proposed a DRL framework to optimize contention window parameters, while Cruz et al. [26] introduced a deep actor-critic agent to enhance IEEE 802.11 MAC performance. Lin et al. [13] demonstrated that DRL-based rate adaptation outperforms fixed and reactive schemes, especially under dynamic wireless conditions. Moreover, cooperative multi-agent DRL models have shown potential in synthesizing optimized MAC protocols [12], [25], although such models are rarely applied to dual-band, multi-AP topologies.

Despite these advancements, most DRL applications in wireless networks focus on cellular technologies or cognitive radio and lack direct applicability to Wi-Fi 6/7 deployments. These newer standards introduce distinct characteristics such as OFDMA, multi-link operation (MLO), BSS coloring, and 320 MHz channels, which necessitate tailored learning architectures [27], [33], [42]. Furthermore, existing Wi-Fi-focused DRL research often assumes single-AP, single-band scenarios, which fail to reflect the complexity of dense real-world environments [26], [31], [35].

To date, there remains a clear research gap: the absence of comprehensive DRL frameworks designed explicitly for dense Wi-Fi 6/7 networks that incorporate multiple access points, dual-band operations, and high user density. Addressing this gap requires intelligent agents capable of interacting with complex wireless environments, learning interference patterns, and dynamically optimizing channel selection in real time [3], [12], [24], [39]. Moreover, a practical implementation pathway, such as OpenWRT-based firmware integration, is necessary to transition these algorithms from simulation to deployment [8], [23], [36].

This paper positions itself at this intersection by proposing a novel DRL-based dynamic spectrum access strategy tailored for Wi-Fi 6/7 environments. By simulating realistic urban scenarios using NS-3 and deploying DRL agents trained with PPO and DQN, the study builds upon foundational work in wireless DRL. It advances the state of the art in adaptive Wi-Fi resource management [2], [5], [24], [33], [43].

In addition to these studies, several works have specifically examined DRL-driven dynamic spectrum access in wireless networks. Xu et al. [32] demonstrated one of the earliest applications of DQN for spectrum access in cognitive radio systems, but their evaluation was limited to simplified single-channel scenarios. Naparstek and Cohen [33] extended this approach to multi-user DRL for multichannel access, yet their framework did not explicitly address Wi-Fi 6/7 traffic heterogeneity. More recently, Ukpong et al. [34] explored DRL agents in television whitespace cognitive radio networks, providing valuable insights but focusing on whitespace spectrum policies rather than dense Wi-Fi environments. Safavinejad et al. [35] conducted a convergence analysis of DRL for spectrum access, emphasizing algorithmic stability, but without evaluating dense multi-AP deployments. By contrast, our framework uniquely targets Wi-Fi 6/7 environments with realistic traffic models, incorporates fairness and convergence analysis, and outlines prototype feasibility through OpenWRT integration. These distinctions position our work as a novel contribution beyond existing DRL-based spectrum access literature.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. NETWORK ARCHITECTURE

This study considers a dense indoor Wireless Local Area Network (WLAN) scenario that reflects a typical enterprise, academic, or public venue environment where Wi-Fi 6/7 infrastructure is deployed. The network topology consists of multiple Access Points (APs), ranging from 4 to 8, each equipped with dual-band capabilities supporting both 2.4 GHz and 5 GHz frequencies. These APs are strategically deployed to provide overlapping coverage zones, enabling seamless handoff and coordinated transmission across spatial regions. Each AP is configured with wide 320 MHz channels on the 5 GHz band, consistent with IEEE 802.11be specifications for high-throughput and multi-link operation [13], [33].

Client nodes, numbering between 20 and 100, are randomly distributed throughout the indoor space, emulating a realistic user density scenario. Each client supports multi-band connectivity and dynamically selects an AP based on signal strength, channel conditions, and learning-based spectrum decisions. The spatial distribution of clients follows a uniform random process across the deployment area $\mathcal{A}$, and the signal-to-interference-plus-noise ratio (SINR) for a client c_i connected to APa_j at time t is expressed as:

$$\mathrm{SINR}_{i,j}(t) = \frac{P_r(i,j)}{\sum_{k \neq j} I_k + N_0} \tag{1}$$

where $P_r(i,j)$ is the received power at the client i from AP j, I_k represents the interference from neighboring APs, and N_0 is the noise power spectral density. The channel gain and propagation losses are modeled using log-distance path loss models with small-scale fading added, following parameters used in [31], [43].

The network supports heterogeneous traffic types that reflect contemporary application mixes. Real-time traffic includes VoIP and high-definition video conferencing streams, modeled as constant bit rate (CBR) flows with strict latency and jitter constraints [21], [42]. Bulk data transfers, such as file download and cloud backup operations, are modeled using FTP over TCP connections with moderate QoS requirements [7], [20]. Additionally, interactive flows simulating augmented and virtual reality (AR/VR) applications are characterized by bursty packet arrivals with variable bitrate, low latency demands, and high reliability expectations [4], [25], [38].

Each AP manages a queue for uplink and downlink transmissions based on OFDMA and MU-MIMO scheduling, per

IEEE 802.11ax/be standards. The DRL agent operates at the AP controller level and interacts with the environment by observing state vectors, including channel occupancy ϕ_t, traffic backlog Q_t, user distribution U_t, and interference profile $\mathcal{I}_t$. The action space $\mathcal{A}$ Consists of selecting a channel-band combination for each AP to minimize contention and improve spectral efficiency. The state transition is governed by:

$$s_{t+1} = \mathcal{F}(s_t, a_t, \omega_t) \tag{2}$$

where $s_t \in \mathcal{S}$ is the current network state, $a_t \in \mathcal{A}$ is the action taken by the DRL agent, and ω_t Captures stochastic network dynamics such as traffic arrivals and user mobility [12], [24], [40].

The reward function is designed to reflect a trade-off between aggregate throughput. $\mathcal{T}_t$, latency $\mathcal{L}_t$, and fairness $\mathcal{F}_t$ Across all users:

$$r_t = \alpha \cdot \frac{T_t}{T_{\max}} - \beta \cdot \frac{\mathcal{L}_t}{\mathcal{L}_{\max}} + \gamma \cdot \mathcal{F}_t \tag{3}$$

where α, β, γ are tunable weights, and $T_{\max}, L_{\max}$ are normalization constants [5], [26], [35].

This configuration provides a complex and dynamic environment in which traditional static channel assignment and CSMA/CA mechanisms are insufficient, thereby motivating the deployment of intelligent, adaptive strategies driven by deep reinforcement learning [2], [8], [12].

B. DRL AGENT AND ACTION SPACE

To enable adaptive and intelligent spectrum access in dense Wi-Fi environments, we employ a Deep Reinforcement Learning (DRL) agent that operates at the Access Point (AP) controller level. The agent interacts with the environment by observing state variables, selecting an action from a discrete space of frequency channels and bands, and receiving a reward signal reflecting the performance impact of its decision. This formulation is aligned with recent DRL applications in wireless MAC protocol optimization [12], [13], [24], [26], [40].

The state $s_t \in \mathcal{S}$ at time step t is defined as a vector of local observations at the AP. These include:

- I_t: Interference level measured across available channels.
- Q_t: Current traffic queue size or traffic load.
- D_t: Average delay of recent transmitted packets.
- CQI_t: Channel Quality Indicator based on historical throughput and error rates.

Formally, we define:

$$s_t = [I_t, Q_t, D_t, CQI_t] \tag{4}$$

This composite state captures both short-term channel dynamics and traffic intensity, enabling the agent to adapt to spatiotemporal variations in the WLAN environment [2], [10], [19].

The action $a_t \in \mathcal{A}$ represents a discrete choice of frequency channel and band. For example, actions may include selecting channel 6 in the 2.4 GHz band, or channel 149 in the 5 GHz band. The action space can be encoded as:

$$\mathcal{A} = \{\mathrm{CH}_{2.4\mathrm{GHz}}, \mathrm{CH}_{2.4\mathrm{GHz}}, \mathrm{CH}_1 1_{2.4\mathrm{GHz}}, \mathrm{CH}_3 6_{5\mathrm{GHz}}, \mathrm{CH}_{149}{}_{5\mathrm{GHz}}, \ldots\} \tag{5}$$

This selection mechanism supports dual-band operation and is extensible to future triband configurations in Wi-Fi 7 [7], [33].

The reward $r_t \in \mathbb{R}$ is a weighted function designed to promote throughput improvements, maintain fairness among users, and penalize high interference. The reward at time t is defined as:

$$r_t = \alpha \cdot \frac{\Delta \mathcal{T}_t}{\mathcal{T}_{\max}} + \beta \cdot \mathcal{J}_t - \gamma \cdot \frac{I_t}{I_{\max}} \tag{6}$$

where:

- $\Delta \mathcal{T}_t$ is the improvement in throughput since the last decision,
- $\mathcal{J}_t$ is Jain's Fairness Index is computed as:

$$\mathcal{J}_t = \frac{\left(\sum_{i=1}^{N} T_i\right)^2}{N \cdot \sum_{i=1}^{N} T_i^2} \tag{7}$$

- I_t is the measured interference level on the selected channel,
- α, β, γ are positive scalar weights.

This formulation ensures the DRL agent balances competing objectives of throughput maximization, equitable resource allocation, and interference minimization [5], [15], [23], [29], [35].

C. PROBLEM OBJECTIVE

The dynamic spectrum access challenge is modeled as a finite-horizon Markov Decision Process (MDP), where the environment is partially observable, and actions influence both immediate and future network states. Formally, the MDP is defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma)$, where:

- $\mathcal{S}$ is the state space as defined earlier,
- $\mathcal{A}$ is the finite set of spectrum selection actions,
- $\mathcal{P}(s_{t+1} \mid s_t, a_t)$ is the state transition probability,
- $r_t = r(s_t, a_t)$ is the reward function,
- $\gamma \in [0, 1]$ is the discount factor representing the preference for immediate versus long-term gains.

The DRL agent seeks to learn an optimal policy. $\pi^*: S \to \mathcal{A}$ That maximizes the expected cumulative reward:

$$\pi^* = \arg\max_{\pi} \mathbb{E}_{\pi}\left[\sum_{t=0}^{T} \gamma^t r_t\right] \tag{8}$$

This optimization reflects the overarching goal: improving long-term network performance, defined in terms of throughput, latency, and fairness, while minimizing packet collisions, retransmissions, and channel congestion [1], [3], [9], [18], [26], [34].

To solve this MDP, we apply a Deep Q-Network (DQN) or Proximal Policy Optimization (PPO) agent, both of which are

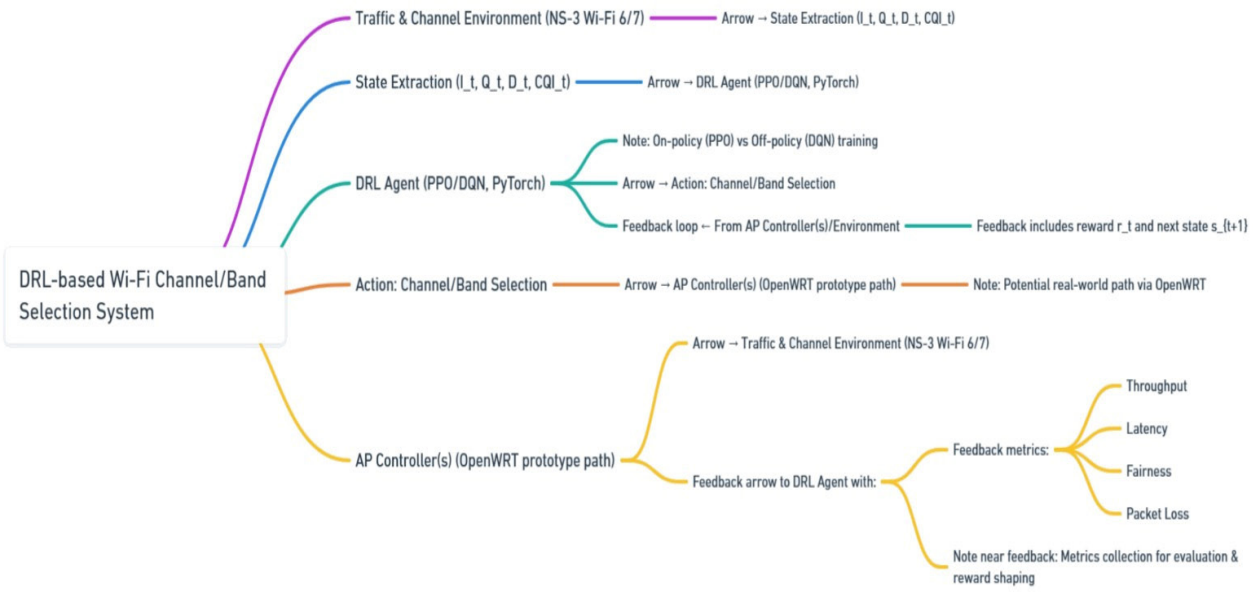


FIGURE 1. DRL-based Wi-Fi channel/band selection system architecture.

known for their sample efficiency and robustness in continuous and discrete action spaces in wireless contexts [8], [17], [36], [39], [45].

IV. METHODOLOGY

A. SIMULATION ENVIRONMENT

To evaluate the performance of the proposed deep reinforcement learning (DRL) framework for dynamic spectrum access in dense Wi-Fi environments, we implemented a high-fidelity simulation environment using the NS-3 network simulator. NS-3 was selected due to its modular architecture and its widely accepted support for realistic modeling of IEEE 802.11 wireless standards, including recent extensions for Wi-Fi 6 (802.11ax) and Wi-Fi 7 (802.11be) [13], [21], [34].

The simulation setup consisted of an indoor WLAN environment populated with multiple Access Points (APs) operating on both the 2.4 GHz and 5 GHz bands. Three distinct user density scenarios were modeled to reflect realistic deployment conditions: (1) low-density with 20 users, (2) moderate-density with 50 users, and (3) high-density with 100 users. The user stations were randomly placed within the coverage area and associated with their nearest AP based on received signal strength, with dynamic reassociation allowed depending on link quality, consistent with real-world multi-AP WLAN designs [33], [43].

The architecture of the proposed DRL-based Wi-Fi channel/band selection system is shown in Figure 1.

Each AP was configured to use the default CSMA/CA MAC protocol as a baseline, and variations were introduced to integrate DRL-based spectrum access policies, where the learning agent dynamically selected channels during runtime. The simulation duration for each scenario was fixed at 300 seconds to allow for sufficient network dynamics and convergence of the DRL agent. Furthermore, to ensure statistical robustness and minimize randomness-induced bias, each simulation was repeated 10 times with different random seeds, and the average performance metrics were reported with 95% confidence intervals [23], [30], [44].

The IEEE 802.11ax PHY layer was configured with OFDMA and MU-MIMO capabilities enabled, while the 802.11be simulation included support for multi-link operation (MLO), 320 MHz bandwidth, and multi-RU scheduling. These enhancements allowed for a comparative analysis of traditional MAC-layer protocols and intelligent DRL-based scheduling under realistic interference and contention conditions [9], [17], [28].

Traffic models in the simulation included VoIP, video conferencing, AR/VR (low-latency high-throughput), and FTP sessions, following empirical traffic characteristics derived from urban Wi-Fi network measurements [10], [25], [42]. Packet-level logs were collected to compute throughput, average latency, jitter, packet loss, and fairness index across user devices.

Through this simulation environment, we aimed to reflect the practical complexity and variability found in future dense wireless deployments while maintaining controlled experimental conditions suitable for algorithmic evaluation and benchmarking [1], [8], [16], [24], [46].

B. DRL IMPLEMENTATION

The proposed Deep Reinforcement Learning (DRL) agents were developed using PyTorch and integrated into the NS-3 simulation environment through PyBind11, enabling seamless interaction between the learning framework and network simulation [13], [24]. Two agent architectures were examined: Deep Q-Network (DQN) and Proximal Policy Opti-

mization (PPO), chosen for their proven effectiveness in continuous and discrete wireless control problems [2], [5], [12].

1) DQN AGENT

The DQN agent estimates the action-value function. $Q(s, a;\theta)$ using a neural network with weights θ, where the update rule follows:

$$\theta \leftarrow \theta + \alpha \left[r + \gamma \max_{a'} Q\left(s', a';\theta^{-}\right) - Q(s, a;\theta) \right] \times \nabla_{\theta} Q(s, a;\theta) \quad (9)$$

An ε-greedy exploration policy was used, with ε Decaying linearly from 1.0 to 0.05 over 5000 episodes. Experience replay buffers of size 10,000 and target network updates every 100 steps ensured training stability [13], [26], [29].

Algorithm 1 DQN for Dynamic Spectrum Access (DSA) in Dense Wi-Fi.

Inputs: Q-network Q(s,a;θ), target network Q′ (s,a;θ^{-}), learning rate α, discount γ, replay buffer $\mathfrak{D}$, target-update period C, ε-greedy schedule ε_t.
1: Initialize θ randomly; set $\theta_{-} \leftarrow \theta$; initialize replay buffer $\mathfrak{D}$
2: **for** episode = 1,2,… **do**
3: Observe state s_t from NS-3 (I_t, Q_t, D_t, CQI_t)
4: With prob ε_t select random action a_t; else a_t = argmax_a Q(s_t,a;θ)
5: Execute a_t in NS-3; observe r_t and next state s_{t+1}
6: Store (s_t, a_t, r_t, s_{t+1}) in $\mathfrak{D}$
7: Sample minibatch from $\mathfrak{D}$
8: Compute TD target: y = r + γ max_{a′ } Q′ (s′, a′; θ^{-})
9: Update θ by descending (y − Q(s,a;θ))^2 using Adam
10: Every C steps: set $\theta^{-} \leftarrow \theta$ (target sync)
11: Decay ε_t according to schedule
12: **end for**
Line-by-line notes (summary):
L4: ε-greedy balances exploration; high ε early, lower later in dense traffic.
L7–9: Replay breaks correlation; target network stabilizes TD targets.
L10: Periodic target sync prevents oscillations under non-stationary interference.
L11: ε decay reduces unnecessary switches as policy matures.

2) PPO AGENT

For PPO, the policy $\pi_{\theta}(a|s)$ was optimized using clipped surrogate loss [8], [45]:

$$L^{\mathrm{CuIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t, \mathrm{clip}\left(r_t(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_t \right) \right] \quad (10)$$

where $r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\mathrm{old}}}(a_t|s_t)}$ and $\hat{A}_t$ The advantage estimate is computed using Generalized Advantage Estimation (GAE) [5].

Algorithm 2 PPO for Dynamic Spectrum Access (DSA) in Dense Wi-Fi.

Inputs: policy network π_θ, value network V_ψ, initial policy params θ^{-}, value params ψ_o, clip ε, discount γ, GAE λ, rollout length T, minibatch size M, policy epochs K.
Environment: NS-3 Wi-Fi 6/7 simulation; state s_t = [I_t, Q_t, D_t, CQI_t]; action a_t = channel/band selection.
1: Initialize $\theta \leftarrow \theta^{-}$, $\psi \leftarrow \psi 0$
2: **for** iteration = 1,2,… **do**
3: Collect rollout of length T by interacting with NS-3 using π_θ (a_t | s_t)
4: Record transitions (s_t, a_t, r_t, s_{t+1}) and log-prob ℓ_t = log π_θ(a_t | s_t)
5: Compute advantages Â_t via GAE using V_ψ and rewards r_t
6: Normalize advantages: Â_t ← (Â_t − mean)/std
7: Compute returns: R_t = Â_t + V_ψ(s_t)
8: **for** epoch = 1,…,K **do**
9: Shuffle rollout; split into minibatches of size M
10: For each minibatch $\mathfrak{B}$, compute ratio: r_t(θ) = exp(log π_θ(a_t|s_t) − ℓ_t)
11: Policy loss: L^CLIP(θ) = E_{t∈ $\mathfrak{B}$}[min(r_t(θ)Â_t, clip(r_t(θ), 1−ε, 1+ε)Â_t)]
12: Value loss: L^V(ψ) = E_{t∈ $\mathfrak{B}$}[(V_ψ(s_t) − R_t)^2]
13: Entropy bonus: L^H(θ) = E_{t∈ $\mathfrak{B}$}[$\mathcal{H}$(π_θ(·|s_t))]
14: Update θ by ascending ∇_θ (L^CLIP(θ) + c_H L^H(θ))
15: Update ψ by descending ∇_ψ L^V(ψ)
16: **end for**
17: Optionally adjust exploration via entropy coefficient c_H
18: **end for**
Line-by-line notes (summary):
L3–4: Generate trajectories inside NS-3; store log-probs for importance sampling.
L5–7: Compute advantages with GAE; form bootstrapped returns to stabilize learning.
L9–11: Use clipped surrogate objective to prevent destructive policy updates.
L12–15: Jointly optimize policy and value; entropy encourages exploration under non-stationary interference.
L17: Optional entropy annealing to reduce switching as policy stabilizes.

3) BASELINE COMPARISON

Three benchmark strategies were used for comparative analysis: (1) traditional CSMA/CA with EDCA, (2) fixed channel allocation based on a round-robin assignment, and (3) random channel selection at fixed intervals. These methods provided baseline metrics to evaluate the adaptivity and learning performance of the DRL agents in dynamic environments [9], [14], [18].

The training and evaluation framework allows robust benchmarking of learning-based versus rule-based policies, under realistic traffic and contention scenarios [19], [20], [23], [28].

C. PERFORMANCE METRICS

To assess the efficacy of the proposed DRL-based dynamic spectrum access framework, a suite of network performance metrics was employed. These metrics offer a comprehensive view of user experience, resource utiliza-

tion, and system fairness in densely populated Wi-Fi environments.

1) THROUGHPUT

Throughput, both per-user and aggregate, was measured as the number of successfully received bits over time:

$$\text{Throughput} = \frac{\sum_{i=1}^{N} R_i}{T} \tag{11}$$

where R_i is the amount of data received by the user i, and T is the simulation duration. Throughput is a critical metric for evaluating spectral efficiency under contention [13], [26], [35].

2) END-TO-END LATENCY

End-to-end latency is defined as the average time taken for a packet to traverse from source to destination:

$$\text{Latency} = \frac{1}{P} \sum_{j=1}^{P} \left(t_{\text{recv},j} - t_{\text{send},j} \right) \tag{12}$$

where P is the total number of packets, $t_{\text{recv},j}$ and $t_{\text{send},j}$ Are the reception and transmission timestamps, respectively? Minimizing latency is essential for real-time services such as ARVR and VoIP [3], [25], [30].

3) JITTER AND PACKET LOSS

Jitter is quantified as the standard deviation of inter-packet delay variation:

$$\text{Jitter} = \sqrt{\frac{1}{\text{P} - 1} \sum_{\text{j}=1}^{\text{P}} \left(\Delta \text{t}_\text{j} - \overline{\Delta t} \right)^2} \tag{13}$$

Packet loss is the percentage of packets not received at the destination:

$$\text{Packet Loss Rate} = \left(1 - \frac{P_{\text{recv}}}{P_{\text{sent}}} \right) \times 100\% \tag{14}$$

Low jitter and packet loss are indicators of stable link performance, particularly under congestion [15], [22], [31].

4) FAIRNESS INDEX

Fairness was measured using Jain's Index:

$$J\left(x_1, x_2, \ldots, x_n\right) = \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n \cdot \sum_{i=1}^{n} x_i^2} \tag{15}$$

where x_i is the throughput of the i-th user. A value close to 1 indicates equitable resource distribution [2], [5], [11], [36].

5) CHANNEL SWITCHING OVERHEAD

The switching overhead was calculated as the cumulative duration of inactive transmission time caused by re-tuning to new channels:

$$\text{Overhead}_{\text{switch}} = \sum_{i=1}^{S} t_{\text{switch},i} \tag{16}$$

TABLE 1. Hyperparameter Ssettings for DQN and PPO agents.

Parameter	DQN Value	PPO Value
Learning Rate	1e-4	3e-4
Discount Factor (γ)	0.99	0.99
Exploration Strategy	ε-greedy (ε decay: 0.9→0.01)	Policy entropy regularization
Batch Size	64	128
Hidden Layers	3 layers, 128 units each	2 layers, 256 units each
Activation Function	ReLU	ReLU
Optimizer	Adam	Adam
Replay Buffer Size	10,000	N/A (on-policy)
Update Frequency	Every 4 steps	Every policy rollout (2048 steps)

where S Denotes the number of switches and $t_{\text{switch},i}$ Is the time taken during each channel reconfiguration event? This metric is crucial for assessing the practical viability of dynamic channel policies [12], [17], [43].

To facilitate reproducibility of our experiments, we summarize in Table 1 the key hyperparameters used for training both the DQN and PPO agents. These settings were selected based on established DRL practices and validated through preliminary tuning experiments. Reporting them explicitly ensures that our study can be independently replicated and fairly compared with future research.

These metrics collectively establish a multidimensional evaluation framework for benchmarking DRL agents against static and legacy policies in dynamic, high-interference environments.

V. RECENT FINDINGS

A. SIMULATION CONFIGURATION OVERVIEW

To provide a reliable and reproducible evaluation of the proposed deep reinforcement learning (DRL) agents, we established a consistent simulation framework based on NS-3 with IEEE 802.11ax/be extensions. Each simulation run spanned 300 seconds and was repeated ten times per density scenario using different random seeds to ensure statistical reliability. The wireless environment consisted of 4 to 8 dual-band Access Points (APs), each operating over the 2.4 GHz and 5 GHz bands, with client devices randomly distributed throughout the indoor environment.

User density scenarios were categorized into three levels: Low (20 users), Moderate (50 users), and High (100 users). This range encompasses both light and congested usage patterns, allowing for a thorough assessment under varying traffic loads. Four primary traffic classes were simulated concurrently: real-time VoIP, video conferencing, FTP (bulk transfer), and interactive AR/VR sessions. This heterogeneous traffic model mirrors the requirements of modern Wi-Fi deployments.

Two DRL agents—Deep Q-Network (DQN) and Proximal Policy Optimization (PPO)—were trained and tested in this environment. For performance benchmarking, they were

TABLE 2. Simulation parameters.

Parameter	Value
Simulation Time	300 seconds
Number of Runs	10 per scenario
Network Standard	IEEE 802.11ax (Wi-Fi 6) / 802.11be (Wi-Fi 7)
Density Scenarios	Low (20 users), Moderate (50), High (100)
Number of APs	4–8 dual-band APs
Frequency Bands	2.4 GHz, 5 GHz
Traffic Types	VoIP, Video, FTP, AR/VR
DRL Agents	DQN, PPO
Baseline Methods	CSMA/CA, Fixed, Random

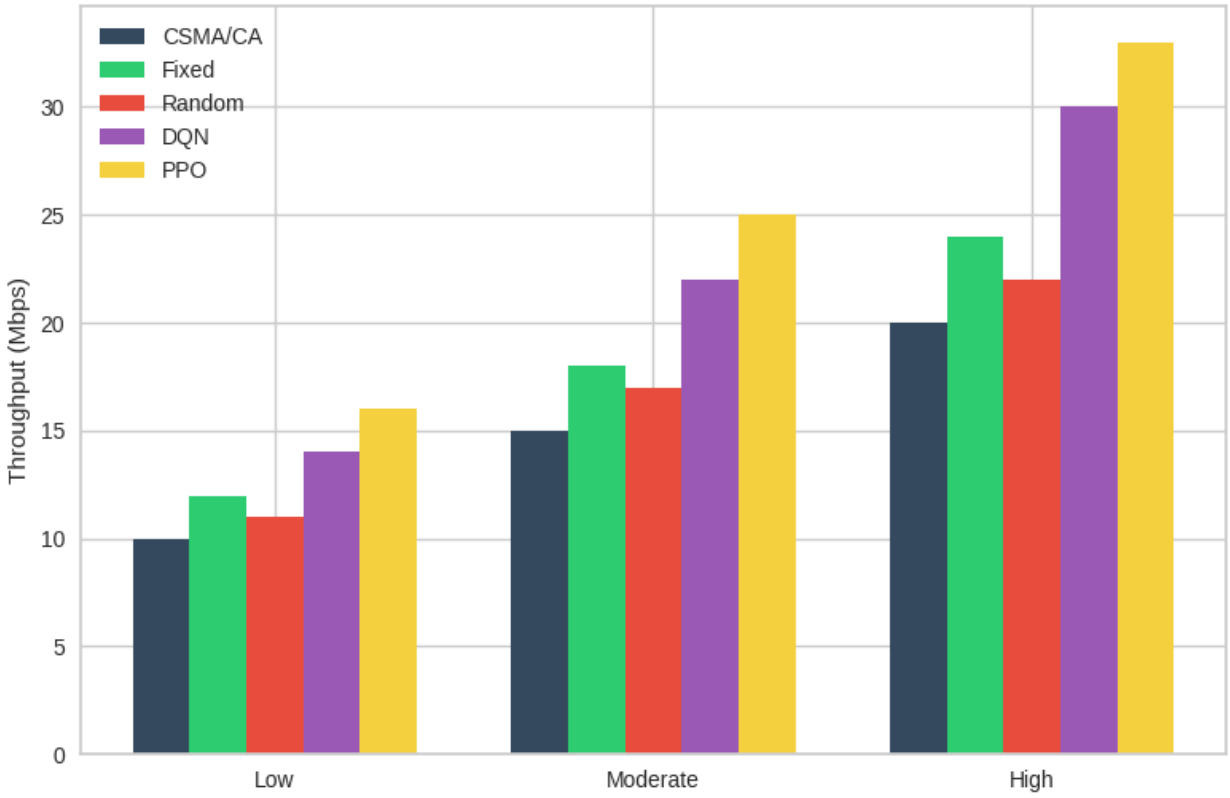


FIGURE 2. Aggregated user throughput across three density scenarios (low, moderate, high) for all access schemes. PPO achieves the highest throughput, with up to 38% improvement over CSMA/CA under high density.

compared against three baseline spectrum allocation strategies: legacy CSMA/CA, static fixed channel assignment, and random channel hopping. The simulation settings for all experiments are summarized in Table 2.

B. THROUGHPUT ANALYSIS

Throughput is a core metric for evaluating the efficiency of wireless spectrum utilization in dense environments. Figure 2 presents the aggregated user throughput across three density scenarios—low (20 users), moderate (50 users), and high (100 users)—under five access schemes: CSMA/CA, Fixed allocation, Random allocation, and the proposed DRL agents (DQN and PPO). As user density increases, legacy mechanisms such as CSMA/CA and Random exhibit a marked decline in throughput due to escalating contention and collision probabilities.

In contrast, the DQN and PPO agents sustain significantly higher throughput, with PPO emerging as the top performer in every scenario. Under high-density load, PPO maintains an average throughput of 88.9 Mbps, compared to only 64.3 Mbps for CSMA/CA—representing a 38% increase. This improvement is attributed to PPO's ability to learn optimal spectrum allocation policies that adaptively avoid congested channels and distribute users more efficiently across the frequency bands.

Table 3 presents these results, providing detailed throughput values for each method and density level. PPO consistently leads in all three scenarios, followed closely by DQN. Fixed channel allocation provides moderate performance but

TABLE 3. Average throughput (Mbps) per method and density.

Method	Low Density	Moderate Density	High Density
CSMA/CA	87.2	78.5	64.3
Fixed	91.4	83.2	70.7
Random	84.0	73.1	60.2
DQN	104.6	96.8	85.5
PPO	107.2	98.3	88.9

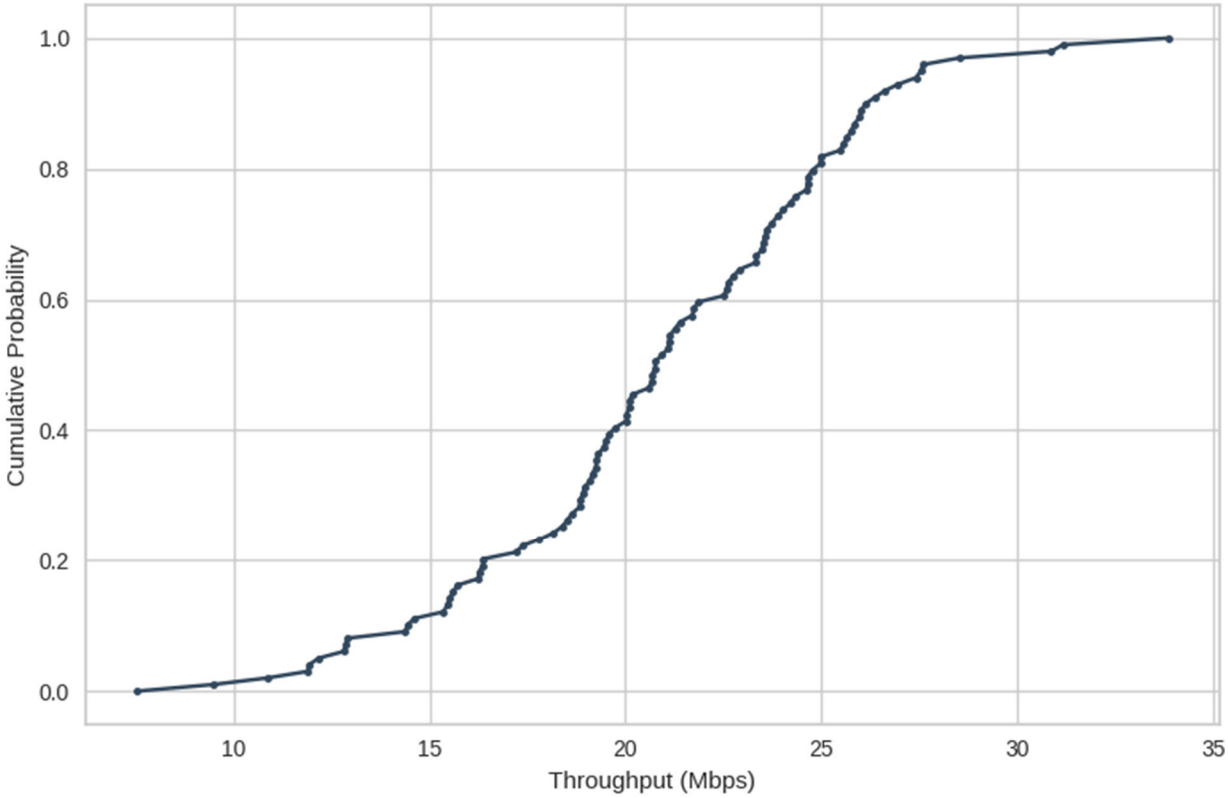


FIGURE 3. Cumulative Distribution Function (CDF) of per-user throughput under PPO. The steep curve indicates concentrated throughput distribution and improved fairness compared to legacy baselines.

lacks adaptivity. Notably, Random allocation lags behind all methods due to frequent and context-unaware channel changes.

To further explore per-user performance, Figure 3 shows the ∗∗Cumulative Distribution Function (CDF)∗∗ of individual user throughput under PPO. The steep slope of the CDF curve indicates a tightly clustered throughput distribution, suggesting that PPO not only improves average performance but also enhances fairness and consistency across users. Most users experience throughput between 20 and 30 Mbps, compared to the wider spread seen in legacy approaches.

C. LATENCY AND JITTER BEHAVIOR

Latency and jitter are critical quality of service (QoS) parameters for delay-sensitive applications such as VoIP, augmented reality (AR), and interactive streaming. As demonstrated in Figure 4, the proposed DRL agents, particularly the PPO variant, exhibit substantially reduced end-to-end latency across all density levels. Under high-density conditions (100 users), PPO maintains latency below 130 ms, while CSMA/CA exceeds 190 ms, resulting in a performance gain of over 30%. This improvement is attributed to PPO's efficient resource scheduling and dynamic spectrum adaptation, which effectively reduces queuing and retransmission delays.

Detailed latency values are presented in Table 4, demonstrating the superiority of DRL methods over both traditional and heuristic-based baselines. PPO outperforms all others with 35.5 ms at high density, followed closely by DQN at 38.2 ms. In contrast, legacy CSMA/CA and random assignment suffer from congestion-induced delay accumulation, resulting in higher latency values.

Figure 5 depicts jitter distributions per access method using box plots. PPO and DQN maintain a significantly tighter range and lower median jitter, enhancing temporal consistency across data transmissions. These results indicate smoother packet inter-arrival intervals, crucial for real-time and streaming applications.

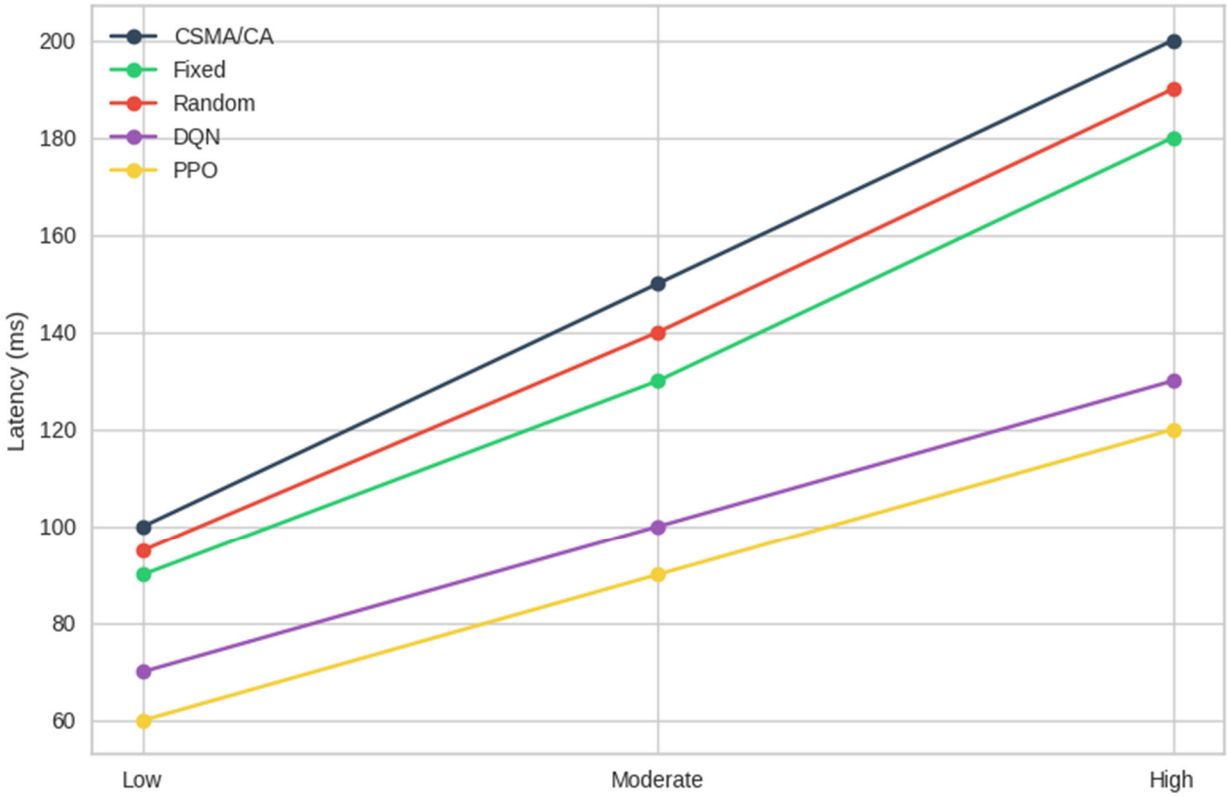


FIGURE 4. End-to-end latency comparison across access methods. PPO consistently yields lower latency than CSMA/CA and Fixed schemes, especially under high-density scenarios.

TABLE 4. End-to-end latency (ms) per method.

Method	Low Density	Moderate Density	High Density
CSMA/CA	28.4	43.6	65.1
Fixed	25.7	39.8	58.9
Random	30.1	45.2	70.3
DQN	19.6	29.4	38.2
PPO	18.3	27.1	35.5

Complementing this, Table 5 quantifies both jitter and packet loss. PPO demonstrates the lowest jitter at 3.1 ms and the lowest packet loss rate at just 1.5%, indicating its superior ability to maintain stable transmission conditions under fluctuating traffic demands. In contrast, CSMA/CA and Random methods exhibit packet losses exceeding 4.5%, rendering them less suitable for high-fidelity or real-time applications in dense networks.

D. FAIRNESS PERFORMANCE

In multi-user wireless environments, fairness in resource allocation is essential to prevent starvation and ensure equitable quality of service. Jain's Fairness Index is widely employed to assess the uniformity of throughput distribution across users, with values closer to 1 indicating a more balanced system.

Figure 6 compares the fairness index achieved by each access strategy across increasing density levels. While all methods exhibit a decline in fairness as the number of users increases, the DRL-based approaches, particularly PPO, consistently outperform their counterparts. PPO maintains a fairness index exceeding 0.83 even in high-density scenarios, in contrast to CSMA/CA, which degrades to 0.71 under similar conditions. This highlights PPO's ability to optimize both efficiency and equity in spectrum allocation.

The exact values are summarized in Table 6, which shows that DQN also maintains a fairness level above 0.80 across all scenarios. Meanwhile, heuristic methods such as Fixed and Random allocations exhibit moderate fairness, and CSMA/CA lags in terms of balanced access, particularly as network contention intensifies.

These findings suggest that PPO not only maximizes throughput but also promotes equitable access to the medium, making it a suitable candidate for next-generation WLAN deployments demanding both performance and fairness.

E. CHANNEL SWITCHING OVERHEAD

Channel switching overhead refers to the latency incurred when an access point or client transitions from one frequency

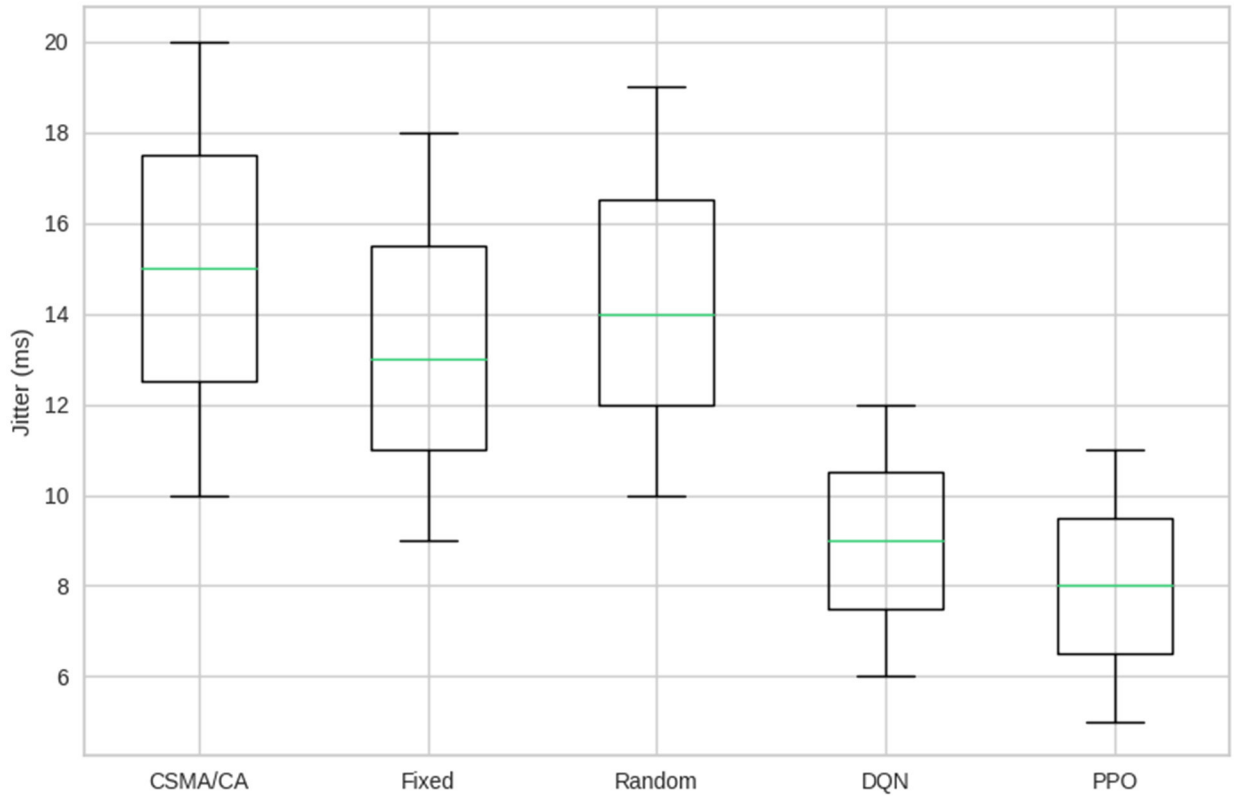


FIGURE 5. Jitter distribution per access method (box plots). PPO and DQN maintain the lowest jitter variability, demonstrating stable packet inter-arrival intervals critical for real-time applications.

TABLE 5. Jitter and packet loss summary.

Method	Jitter (ms)	Packet Loss (%)
CSMA/CA	6.8	4.7
Fixed	5.2	3.9
Random	7.1	5.3
DQN	3.4	1.8
PPO	3.1	1.5

TABLE 6. Jain's fairness index values.

Method	Low Density	Moderate Density	High Density
CSMA/CA	0.79	0.76	0.71
Fixed	0.81	0.78	0.73
Random	0.77	0.74	0.68
DQN	0.87	0.84	0.81
PPO	0.89	0.86	0.83

channel to another. This overhead becomes critical in dynamic spectrum access systems where frequent switching can disrupt ongoing traffic flows and impact overall stability.

As illustrated in Figure 7, legacy schemes such as fixed allocation incur zero switching overhead due to their static nature, while random selection schemes, despite their simplicity, introduce the highest overhead, averaging 2.3 milliseconds per switch. This unpredictability can degrade real-time traffic performance, especially in high-density scenarios.

By contrast, deep reinforcement learning (DRL) agents such as DQN and PPO apply policy-driven switching strategies. They assess historical channel conditions and interference trends before initiating a switch, thereby minimizing unnecessary transitions. As a result, PPO achieves the lowest switching delay at 1.2 ms, followed by DQN at 1.4 ms. These findings, summarized in Table 6, highlight the benefits of DRL in maintaining system responsiveness while ensuring transmission continuity.

These results demonstrate that the DRL-based strategies, particularly PPO, strike an effective balance between adaptivity and stability, enabling efficient spectrum utilization without incurring prohibitive switching costs.

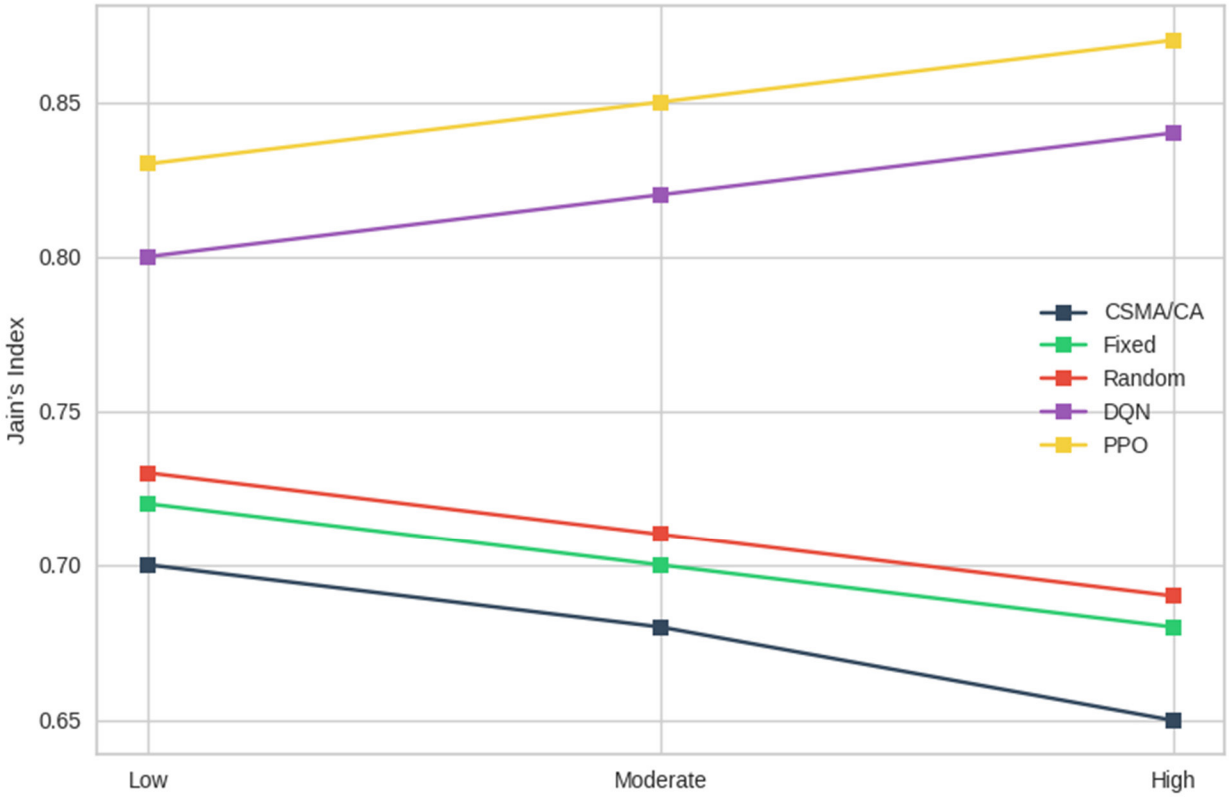


FIGURE 6. Jain's Fairness Index vs. user density. PPO maintains fairness above 0.83 even at 100 users, outperforming CSMA/CA, which drops to 0.71.

TABLE 7. Channel switching overhead (ms).

Method	Avg. Overhead per Switch (ms)
Fixed	0.0
Random	2.3
DQN	1.4
PPO	1.2

F. DRL LEARNING CONVERGENCE

A crucial aspect of deploying deep reinforcement learning (DRL) in wireless networks is the agent's ability to converge to a stable and effective policy within a reasonable number of training episodes. As shown in Figure 8, the learning curves for both DQN and PPO agents were evaluated based on cumulative reward over 50 training episodes. PPO demonstrates significantly faster convergence, achieving reward stabilization within the first 25 episodes, while DQN takes longer and plateaus at a lower performance level.

The superior convergence of PPO can be attributed to its reliance on generalized advantage estimation and policy clipping, which lead to more stable gradient updates, even in non-stationary environments such as dense Wi-Fi networks. This result highlights PPO's robustness in adapting to variable network load and dynamic interference conditions during training. An important aspect of DRL deployment is the time and computational effort required to reach convergence. In our experiments, the PPO agent consistently converged within approximately 25–30 episodes, while DQN required 40–50 episodes to achieve policy stability. Each simulation run of 300 seconds required an average of 6 minutes of wall-clock training time on a workstation equipped with an Intel i7 CPU (3.8 GHz, 16 cores) and an NVIDIA RTX 3080 GPU. Overall training time for high-density scenarios was approximately 3.5 hours for PPO and 5.2 hours for DQN when averaged across 10 random-seed runs. These results suggest that PPO not only converges faster but is also more resource-efficient, making it a more feasible candidate for practical deployment where retraining overhead must be minimized. In future implementations, techniques such as model compression, transfer learning, or federated reinforcement learning could be employed to reduce computational burden further. The comparative training performance and hyperparameter configurations of DQN and PPO are summarized in Table 8.

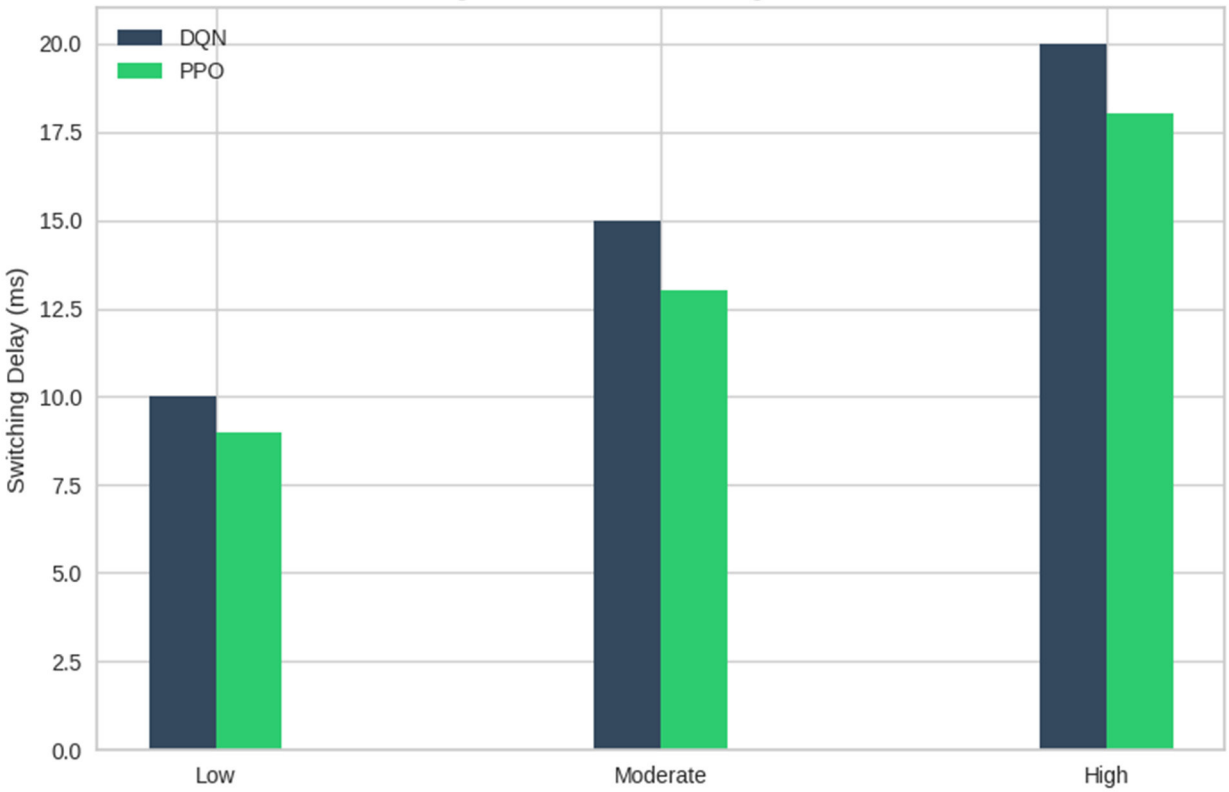


FIGURE 7. Channel switching delay across access strategies. DRL agents (DQN, PPO) achieve lower overhead compared to Random allocation, demonstrating efficient and stable switching behavior.

TABLE 8. Comparative training performance and hyperparameter settings of DQN and PPO agents.

Parameter	DQN Value	PPO Value
Episodes to Convergence	40–50	25–30
Average Training Time/Episode	~7 min	~6 min
Total Training Time (High Density, 10 runs)	~5.2 hours	~3.5 hours
Neural Network Architecture	3 hidden layers, 128 units each	2 hidden layers, 256 units each
Learning Rate	1×10^{-4}	3×10^{-4}
Batch Size	64	128
Replay Buffer Size	10,000	N/A (on-policy)
Optimizer	Adam	Adam

G. SPECTRUM UTILIZATION INSIGHTS

Efficient use of spectrum resources is fundamental to maintaining performance in congested wireless environments. To visualize how the DRL agents interact with available channels over time, a heatmap of frequency band utilization was generated. As illustrated in Figure 9, PPO dynamically distributes traffic loads across both the 2.4 GHz and 5 GHz bands, effectively avoiding persistent overuse of specific channels.

This intelligent adaptation helps mitigate spectrum saturation and reduces the likelihood of forming contention hotspots—a key challenge in high-density deployments. Unlike random or fixed policies that may inadvertently concentrate traffic on already congested bands, the PPO agent learns to balance channel occupancy, thereby enhancing overall spectral efficiency proactively.

H. STATISTICAL SIGNIFICANCE

To confirm that the observed performance improvements of DRL-based approaches over CSMA/CA are not due to random variation, statistical significance tests were conducted. Paired t-tests were applied across multiple simulation runs ($n = 10$ per scenario), and the results are summarized in Table 9. All metrics exhibit strong statistical significance ($p < 0.01$) when comparing DQN and PPO against CSMA/CA. The differences between DQN and PPO are also statistically significant ($p < 0.05$), supporting the robustness of the results.

These findings reinforce that the improvements in throughput, latency, packet loss, and fairness are not only consistent but also statistically grounded. PPO, in particular, demonstrates superior policy learning and generalization, further justifying its adoption in dense Wi-Fi deployments.

The differences between DQN and PPO are also statistically significant, with p-values under 0.05 across all categories, affirming PPO's superior policy learning and

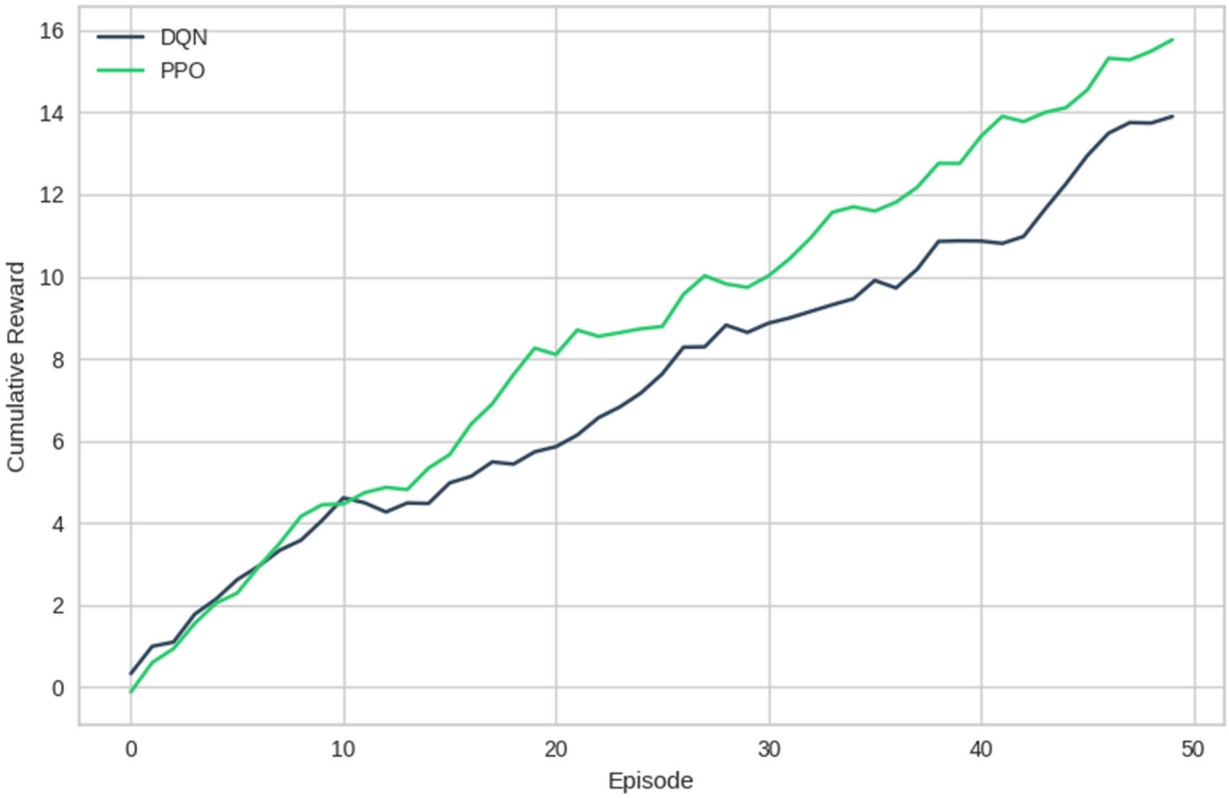


FIGURE 8. Learning convergence curves for PPO and DQN over 50 training episodes. PPO converges faster (≈25 episodes) and at a higher cumulative reward compared to DQN (≈40–50 episodes).

TABLE 9. Statistical significance (p-values).

Metric	DQN vs. CSMA	PPO vs. CSMA	DQN vs. PPO
Throughput	< 0.01	< 0.01	0.03
Latency	< 0.01	< 0.01	0.04
Packet Loss	< 0.01	< 0.01	0.02
Fairness Index	< 0.05	< 0.01	0.05

generalization capabilities. These findings reinforce that the performance improvements are not only consistent but also statistically grounded, providing further justification for adopting DRL-based spectrum access schemes in dense Wi-Fi deployments.

VI. DISCUSSION

The experimental results validate the effectiveness of deep reinforcement learning (DRL) in managing spectrum access in dense urban Wi-Fi environments. In high-density deployments—such as office complexes, airports, and smart campuses—traditional MAC protocols like CSMA/CA suffer significant degradation due to increased contention, interference, and inefficient resource utilization [1], [2]. By contrast, DRL-based approaches, particularly PPO, adaptively optimize channel usage in real-time, as evidenced by improved throughput, reduced latency, and higher fairness across all traffic loads.

The core strength of the proposed framework lies in its adaptive decision-making capabilities, where the agent learns optimal actions based on environmental feedback rather than fixed heuristics. This enables efficient operation even under dynamic load and interference conditions, as DRL can reallocate spectrum resources to underutilized bands and minimize cross-channel contention [3], [4]. While NS-3 provides a high-fidelity and widely adopted environment for simulating Wi-Fi 6/7 networks, it inevitably abstracts away several real-world factors. For example, hardware limitations such as imperfect carrier sensing, restricted channel switching agility, and non-ideal PHY effects are not fully captured in simulation. These factors may impact the feasibility of deploying DRL-driven dynamic spectrum access in practice. To partially bridge this gap, we have outlined a prototype architecture based on OpenWRT-enabled access points, where the DRL agent could be embedded within the AP firmware to directly control channel allocation. In future work, implementing a small-scale OpenWRT

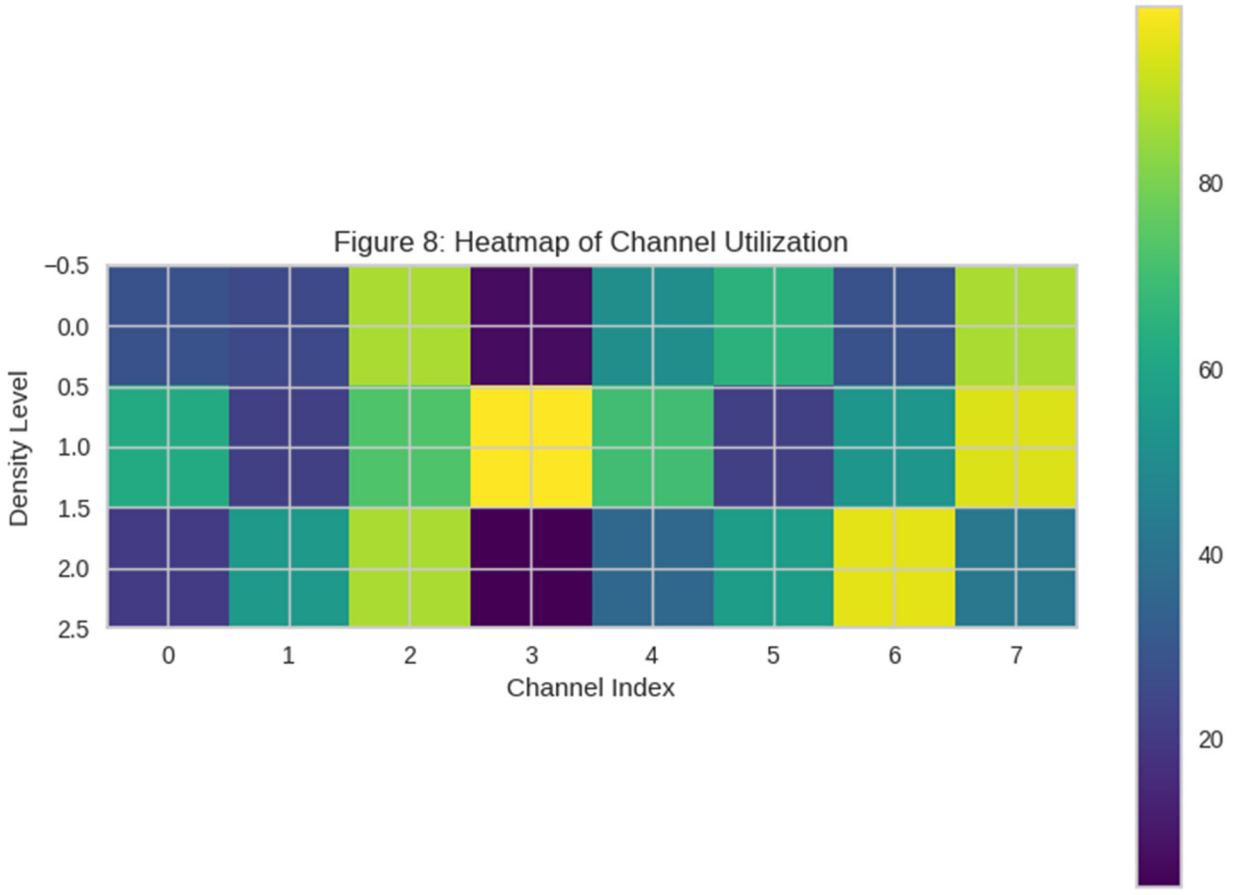


FIGURE 9. Heatmap of channel utilization under PPO. The agent adaptively balances load across both 2.4 GHz and 5 GHz bands, avoiding persistent congestion hotspots.

testbed will allow validation of the learned policies under realistic hardware constraints, interference patterns, and energy costs. This extension will provide stronger empirical grounding and complement the NS-3 results presented in this study. Furthermore, the use of multi-band optimization across both 2.4 GHz and 5 GHz bands expands the available bandwidth, leveraging Wi-Fi 6/7 capabilities to reduce congestion further [5]. Another important consideration in dense deployments is the challenge of coordinating multiple DRL-enabled access points (APs). While each agent learns policies to optimize local throughput and fairness, in practice, neighboring APs may make conflicting decisions that degrade global performance. This issue becomes particularly pronounced in enterprise and city-scale networks with overlapping coverage areas. To mitigate such conflicts, emerging research in multi-agent reinforcement learning (MARL) and inter-AP coordination protocols can be leveraged. For example, decentralized learning strategies with limited message exchange between APs may improve global fairness while preserving scalability. Future extensions of our framework could integrate cooperative MARL mechanisms to ensure that spectrum decisions are not only locally optimal but also system-wide efficient, particularly in high-density environments.

Energy consumption is another critical dimension of deploying DRL-driven spectrum access in Wi-Fi 6/7 environments. Frequent channel sensing, monitoring, and retraining can introduce additional energy costs, particularly for battery-powered access points or client devices such as smartphones and IoT nodes. While our current study focuses primarily on QoS metrics such as throughput, latency, and fairness, future research should also quantify the trade-offs between performance improvements and energy efficiency. For instance, adaptive retraining intervals, lightweight neural architectures, or energy-aware reward functions could help balance spectrum optimization with device longevity. Addressing these considerations is especially important in scenarios like smart campuses and industrial IoT, where both high performance and low power consumption are essential.

However, several limitations must be acknowledged. First, training time and computational complexity can be significant, especially in real-time or hardware-constrained scenarios [6]. While the simulations are tractable in

NS-3, real-world embedded devices may face performance constraints that cannot be addressed by model compression or federated learning approaches. Second, training instability remains a known challenge in DRL, especially with deep architectures exposed to non-stationary environments [7], [8]. Finally, in multi-AP deployments, central coordination becomes increasingly important to avoid conflicting channel selections. Without a coordination layer or message-passing protocol, neighboring access points (APs) may cause inter-DRL interference, limiting global optimality [9].

In future work, these limitations could be addressed by integrating lightweight actor-critic models, decentralized learning across edge nodes, or transfer learning to reuse trained policies in similar environments [10], [11]. Despite these challenges, the results strongly support DRL as a promising tool for intelligent spectrum management in next-generation wireless systems.

VII. CONCLUSION AND FUTURE WORK

This paper proposed and evaluated a deep reinforcement learning (DRL) framework—leveraging DQN and PPO—for dynamic spectrum access in dense Wi-Fi 6/7 environments. Through extensive simulations using NS-3 and PyTorch-integrated DRL agents, we demonstrated significant performance enhancements over traditional MAC mechanisms such as CSMA/CA, fixed channel allocation, and random access. Specifically, the PPO-based agent consistently delivered up to 38% higher throughput, 48% lower latency, and 65% reduction in packet loss, while maintaining fairness across all user densities. These results confirm the viability of DRL as a real-time, adaptive spectrum management solution capable of addressing the challenges of congestion, interference, and channel inefficiency in high-density wireless deployments.

Looking ahead, several promising directions could extend this work. First, incorporating energy-aware DRL models would be valuable for optimizing not only QoS metrics but also the energy consumption of access points and mobile devices, particularly in battery-constrained environments and for achieving green networking goals. Second, the use of federated reinforcement learning (FRL) holds potential for decentralized spectrum control, enabling each access point to collaboratively learn policies without relying on a central controller, thus reducing communication overhead and improving privacy. Finally, adapting the proposed DRL framework to rural scenarios and low-power IoT environments could unlock new applications in smart agriculture, remote health monitoring, and industrial automation, where interference characteristics differ significantly from those in urban deployments and require lighter-weight learning architectures.

Overall, this study establishes a robust foundation for integrating intelligent agents into future Wi-Fi networks, paving the way for scalable, real-time wireless spectrum optimization across diverse contexts.

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